

Lab 03 Propositional Equivalences

Objective

Solving exercises from the textbook in chapter 1.3

Current Lab Learning Outcomes (LLO)

By completion of the lab, the students should be able to:

1. Understand propositional equivalences and.
2. prove the propositional equivalences using rules

Lab Requirements

Students allowed using their lecture notes in the lab in order to solve the exercises.

Lab Assessment

- 1- Divide students to groups and let them to solve the given example.
- 2- Discuss the answers with the groups and write on board the optimal solution.

Lab Description

1. Show that $\neg(p \vee \neg(p \wedge q))$ is contradiction using rules.
2. show that $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$ is a tautology using rules
3. Show that $[p \wedge (p \rightarrow q)] \rightarrow q$ is a tautology using rules
4. Show that $[\neg p \wedge (p \vee q)] \rightarrow q$ is a tautology using rules.
5. Show that $\neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$ are logically equivalent using rules.
6. Show that $\neg p \leftrightarrow q$ and $p \leftrightarrow \neg q$ are logically equivalent using rules
7. Determine whether $\neg(p \vee (\neg p \wedge q))$ and $(\neg p \wedge \neg q)$ equivalent using rules?